

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA0713
NETWORKS

COURSE NAME: COMPUTER

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Ahamed Kaif S(192321128) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Ahamed Kaif S

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Suppose we want to transmit the message 11100011 and protect it from errors using the CRC polynomial X^3+1 .

- i.) Use polynomial long division to determine the message that should be transmitted.
- ii.) Suppose the leftmost bit of the message is inverted due to noise on the transmission link. What is the result of the receiver's CRC calculation? How does the receiver know that an error has occurred?

CRC Calculation

Given:

- Message = **11100011**
- Generator Polynomial = x^3+1x^3+1
- Generator Bits = **1001**

The message is appended with three zeros and divided by the generator polynomial using modulo-2 division. The CRC remainder obtained is:

CRC Remainder = 100

Hence, the transmitted message is:

Transmitted Codeword = 11100011100

Error Detection

Assume the **leftmost bit is inverted** during transmission due to noise.

- **Transmitted Codeword:** 11100011100
- **Received Codeword:** 01100011100

The receiver performs the CRC check by dividing the received codeword by the same generator polynomial (1001).

Receiver's CRC Remainder = 111

Since the remainder is **not equal to 000**, the received codeword is **not exactly divisible** by the generator polynomial. Therefore, the receiver concludes that an error has occurred during transmission.

2. a) A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?
- b) A network has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal- to-noise ratio is 3162. Find the channel capacity

a) Throughput of the Network

Given:

- Bandwidth = **10 Mbps**
- Frames transmitted = **12,000 frames/minute**
- Average frame size = **10,000 bits**

Calculation:

Frames per second:

$$\frac{12000}{60} = 200 \text{ frames/second}$$

Throughput:

$$200 \times 10000 = 2,000,000 \text{ bits/second}$$

Throughput = 2 Mbps

b) Channel Capacity

Given:

- Bandwidth, $B = 3000 \text{ Hz}$
- Signal-to-Noise Ratio (SNR) = 3162

Using the Shannon Capacity Formula:

$$C = B \log_2(1 + \text{SNR})$$

$$C = 3000 \log_2(1 + 3162)$$

Since,

$$\log_2(3163) \approx 11.63 \quad | \quad \log_2(3163) \approx 11.63 \quad | \quad \log_2(3163) \approx 11.63 \quad | \quad C = 3000 \times 11.63 \quad | \quad C = 3000 \times 11.63 \quad | \quad C \approx 34,890 \text{ bps} \quad | \quad C \approx 34,890 \text{ bps}$$

Channel Capacity ≈ 34.9 kbps

3. A multispecialty hospital uses wireless patient-monitoring devices to continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the data is transmitted over a wireless channel, electromagnetic interference occasionally changes one bit during transmission. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital implements the **Hamming (12,8)** error-correcting code.

Hamming (12,8) Error-Correcting Code – Hospital Patient Monitoring

In a multispecialty hospital, wireless patient-monitoring devices continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the transmission occurs over a wireless channel, electromagnetic interference may occasionally change a single bit during transmission. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital uses the **Hamming (12,8) error-correcting code**.

Working of Hamming (12,8):

- The original **8-bit patient data** is encoded into a **12-bit codeword** by adding **4 parity bits**.
- During transmission, if **one bit is altered** due to interference, the receiver recalculates the parity bits and determines the **error syndrome**.
- The syndrome identifies the exact position of the erroneous bit.
- The receiver flips the incorrect bit to restore the original data automatically.
- The corrected patient information is then displayed at the central monitoring station without requiring retransmission.

Advantages in the Hospital Scenario:

- Detects and corrects **single-bit errors**.
- Ensures reliable transmission of critical patient data.
- Eliminates delays caused by retransmission.
- Improves the accuracy of real-time patient monitoring.
- Enhances patient safety by providing doctors with error-free medical information.

4. a) A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

b) A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

a) IPv4 Subnetting

Given:

- Network Address: **172.16.10.0/24**
- Subnet Mask: **/27 (255.255.255.224)**

Solution

- **Number of Subnets**

$$\text{Borrowed bits} = 27 - 24 = 3$$

$$\text{Number of subnets} = 2^3 = 8$$

- **Usable Hosts per Subnet**

$$\text{Host bits} = 32 - 27 = 5$$

$$\text{Total addresses per subnet} = 2^5 = 32$$

$$\text{Usable hosts} = 32 - 2 = 30$$

- **Subnet Ranges**

Subnet	Network Address	Host Range	Broadcast Address
1	172.16.10.0	172.16.10.1 – 172.16.10.30	172.16.10.31
2	172.16.10.32	172.16.10.33 – 172.16.10.62	172.16.10.63

3	172.16.10.64	172.16.10.65 – 172.16.10.94	172.16.10.95
4	172.16.10.96	172.16.10.97 – 172.16.10.126	172.16.10.127
5	172.16.10.128	172.16.10.129 – 172.16.10.158	172.16.10.159
6	172.16.10.160	172.16.10.161 – 172.16.10.190	172.16.10.191
7	172.16.10.192	172.16.10.193 – 172.16.10.222	172.16.10.223
8	172.16.10.224	172.16.10.225 – 172.16.10.254	172.16.10.255

IP Address: 172.16.10.78

- **Subnet:** 172.16.10.64/27
- **Network Address:** 172.16.10.64
- **Broadcast Address:** 172.16.10.95

IP Address: 172.16.10.95

- **172.16.10.95 is the broadcast address** of the subnet **172.16.10.64/27**.
- Therefore, it belongs to the same subnet but **cannot be assigned to a host**.

Final Answers

- **Total Subnets: 8**
- **Usable Hosts per Subnet: 30**
- **172.16.10.78 belongs to: 172.16.10.64/27**
- **Network Address: 172.16.10.64**
- **Broadcast Address: 172.16.10.95**
- **172.16.10.95:** Same subnet, but it is the **broadcast address**.

b) IPv6 Addressing

Given IPv6 Address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Compressed Form

Remove leading zeros and replace consecutive zero groups with ::

Compressed Address:

2001:db8::ff00:42:8329

Expanded Form

Expanding back to full notation:

2001:0db8:0000:0000:0000:ff00:0042:8329

Network Prefix (/64)

With a **/64 prefix**, the first 64 bits represent the network.

Network Prefix:

2001:0db8:0000:0000::/64

Total Number of Host Addresses

Host bits = $128 - 64 = 64$

Total possible host addresses:

$$2^{64} = 18,446,744,073,709,551,616 \quad =$$

≈ 18.4 quintillion host addresses

Final Answers

- **Compressed Address:** 2001:db8::ff00:42:8329
- **Expanded Address:** 2001:0db8:0000:0000:0000:ff00:0042:8329
- **Network Prefix (/64):** 2001:0db8:0000:0000::/64
- **Total Host Addresses:** $2^{64} = 18,446,744,073,709,551,616$ (≈ 18.4 quintillion)

5. A university is conducting an online semester examination for **8,000 students** simultaneously. At exactly **10:00 AM**, the examination server begins transmitting encrypted question papers to all students. The university server is connected through a **10 Gbps highspeed network**, while students access the examination portal using different internet connections such as **fiber broadband (100 Mbps), home Wi-Fi (20 Mbps), and mobile networks (5–10 Mbps)**. During the first few minutes of the examination, many students with slower internet connections report incomplete downloads, repeated retransmissions, delayed acknowledgments, and temporary freezing of the examination portal. Network administrators observe that the server continues transmitting data at a high rate even though several student devices are unable to receive and

process packets at the same speed. This results in receiver buffer overflow, unnecessary retransmissions, increased network traffic, and delays in delivering the question papers. **Analyze** the above scenario and identify the primary networking problem. Explain how the mismatch between the sender's transmission speed and the receiver's processing capability affects communication.

The primary networking problem in this scenario is **Flow Control**.

The university examination server is connected to a **10 Gbps** network and transmits encrypted question papers at a very high speed. However, students access the examination portal using slower internet connections such as **100 Mbps fiber broadband, 20 Mbps Wi-Fi, and 5–10 Mbps mobile networks**. This creates a mismatch between the sender's transmission speed and the receivers' ability to receive and process data.

Because the server sends data faster than many student devices can handle, the receivers' buffers become full, leading to **buffer overflow**. As a result, some packets are lost, causing **retransmissions, delayed acknowledgments (ACKs)**, and increased network congestion. Students experience incomplete downloads, temporary freezing of the examination portal, and delays in receiving the question papers.

Flow control prevents this problem by regulating the rate at which the sender transmits data according to the receiver's available buffer space and processing capability. Protocols such as **TCP** use mechanisms like the **sliding window** to ensure that the sender does not overwhelm the receiver. This reduces packet loss, avoids unnecessary retransmissions, improves network efficiency, and ensures reliable delivery of the examination data.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem

Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation